

Supplementary Material

Pauli Correlation Encoding for Distribution Grid Expansion Planning

Team eQoSystem, E.ON problem statement. Prior results in full, with the result file behind each table in the public repository <https://github.com/AshrafBoussahi/eQoSystem-EON-GridExpansion>.

Every number below was produced before this proposal was written and is reproduced by the code in the repository; nothing here is projected. Tables are grouped by the claim they support in the main text.

Appendix A. The three-setting property, measured on a 108-qubit processor

The claim in Section 2.2 of the proposal is operational: the number of distinct circuits the device must execute is three, whatever the decision count. We took two problem sizes to the Rigetti Cepheus-1-108Q through the direct provider and submitted each in exactly three measurement settings at 1,024 shots. Table A1 is the campaign record; Figure A1 shows the routed circuit resources and the Bell control that validates the read-out pipeline on every edge used.

Table A1: Measurement campaign on the Rigetti Cepheus-1-108Q. Both problem sizes are read from three device jobs and the same 3,072 shots. Magnitude retention is a least-squares slope against exact simulation of the same circuit; chance-level sign agreement is 50%. The last two rows are the constraint that sets the depth bar in Section 3 of the proposal.

Quantity	$m = 45, n = 6$	$m = 105, n = 7$
Correlation order k	2	3
Generated gates	59	68
Two-qubit gates after routing	32.3	40.7
Depth after routing	58	81
Device jobs	3	3
Shots consumed	3,072	3,072
Correlators recovered	45 of 45	105 of 105
Offline check, emitted QASM against trainer state	fidelity 1.000000000000	correlators to 3×10^{-12}
Bell control on device edges, signs correct	24 of 24 (8 edges, three correlators each)	
Bell control, mean magnitude $\langle ZZ \rangle, \langle XX \rangle, \langle YY \rangle$	0.82, 0.81, 0.74 (none below 0.64)	
Read-out error per qubit	6.1%	6.0%
Correlator magnitude retained at 59 to 68 gates	5.1%	2.5%

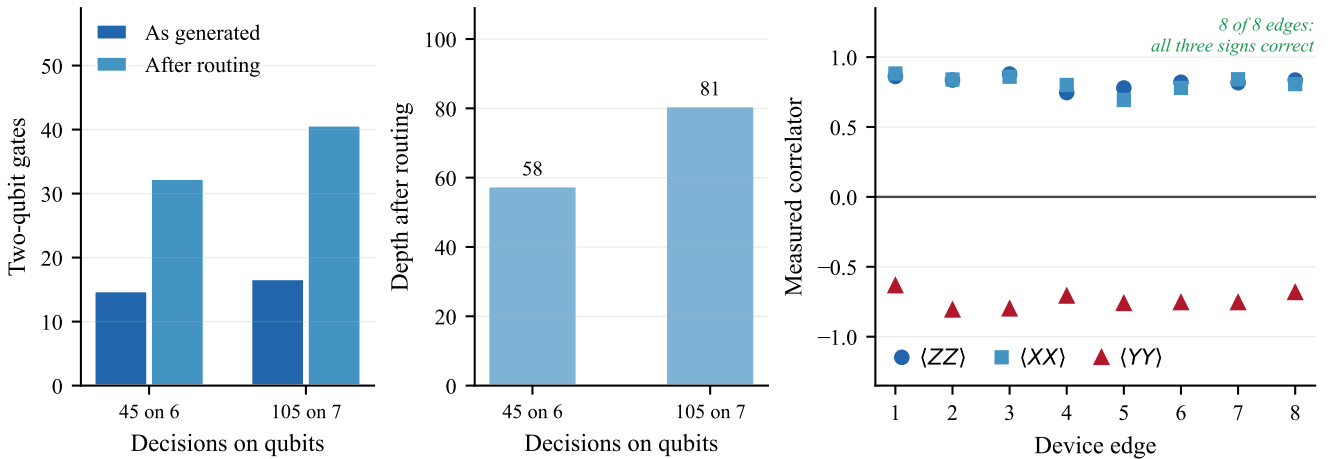


Figure A1: Circuit resources of the generated circuits before and after routing onto the device lattice (left, centre), and the two-qubit Bell control on the eight nearest-neighbour edges used (right): every edge reproduces the expected sign of all three correlators.

Appendix B. Discrete native-gate PCE circuits against variational PCE

Section 2.3 of the proposal states that gradient-free discrete circuits lead the variational method at low quantum budget. Table A2 is the full record at three sizes on random 3-regular and Erdős-Rényi graphs with certified (mixed-integer) or 30-restart best-known values; the variational arm is the brickwork ansatz of the original PCE paper trained by Adam with parameter-shift gradients, charged $2N_p + 1$ circuit executions per step as a real device would be. Every entry is a mean raw approximation ratio (no classical post-processing) over four instances and two runs (three to seven runs at $m = 858$).

Table A2: Mean raw approximation ratio at matched circuit-execution budgets. The discrete search leads at 5,000 and 20,000 executions at every size; the variational method overtakes above 10^4 to 10^5 and reaches its final value at 15 to 30 times the budget. The last column is the median correlator magnitude, which governs shot robustness.

m (n)	Method	@5k	@20k	@100k	Final (budget)	Median $ c $
60 (6)	Discrete PCE search	0.885	0.897	0.897	0.897 (20k)	0.11
60 (6)	Variational PCE, 8 layers	0.811	0.862	0.922	0.959 (414k)	0.24
60 (6)	Random circuits	0.778	0.792		0.792 (20k)	0.06
252 (9)	Discrete PCE search	0.734	0.743	0.743	0.743 (20k)	0.03
252 (9)	Variational PCE, 8 layers	0.665	0.721	0.801	0.853 (639k)	0.13
252 (9)	Random circuits	0.670	0.682		0.682 (20k)	0.03
858 (13)	Discrete PCE search	0.664	0.673	0.673	0.673 (20k)	0.02
858 (13)	Variational PCE, 8 layers	0.615	0.650	0.689	0.738 (676k)	0.11
858 (13)	Random circuits	0.609	0.616		0.616 (20k)	0.02

The smoothed term as a shot-robustness device. The reward used by the circuit writer combines the exact decoded objective with the tanh surrogate of the original PCE paper. At $m = 252$ and 20,000 evaluations (Table A3) the exact objective alone produces circuits whose correlators sit near zero and lose 13 points when read from 1,000 shots; the surrogate restores the magnitude and most of the shot-decoded quality. This is why the hardware campaign trains with the surrogate on.

Table A3: Reward composition against shot-decoded quality, $m = 252$, 20,000 evaluations, one instance. “Relaxed” is the tanh surrogate.

Reward	Exact ratio	1,000 shots	4,000 shots	Median $ c $
Exact objective only	0.724	0.591	0.595	0.019
Exact + $4 \times$ relaxed	0.756	0.688	0.707	0.072
Relaxed only	0.716	0.698	0.711	0.109

Appendix C. Grid results behind Sections 2.1 and 2.4

Benchmark frame. Table A4 is the classical frame around our transmission-siting family: 80 instances on four IEEE systems with certified optima, every arm at a matched 10,000-evaluation budget. We report the arms that beat us as well as those we beat; the exact-match target in the proposal (50% on $m \leq 105$) is set against the 48.8% here.

Table A4: Storage-siting benchmark, 80 instances with certified optima, every arm at 10,000 evaluations. “Exact” is the share reaching the certified optimum; “gap” the mean distance from it; the last column pairs each arm against the correlation solver on identical instances (Wilcoxon signed-rank).

Arm	Exact	Gap	Paired against ours
Exact MIP (reference)	100%	0.000	3.9 ms mean solve time
Simulated annealing	65.0%	0.588	better, $p = 4.7 \times 10^{-4}$
Unconstrained-input control	55.0%	0.879	tied, $p = 0.402$
Tabu search	53.8%	0.712	better, $p = 0.033$
Correlation solver (ours)	48.8%	0.984	
Greedy, 20 restarts	17.5%	7.977	worse, $p = 5.7 \times 10^{-12}$
Greedy, 1 restart	12.5%	12.018	worse, $p = 3.6 \times 10^{-13}$

N-1 screening of decoded plans. Every decoded plan is screened against the complete single-outage set with a security-constrained DC optimal power flow that may shed load at a value of lost load of 1,000 \$/MWh, with and without the plan on identical outages (Figure A2, Table A5). This is the screen the PoC applies to every line plan, so that outage risk, one of the three trade-offs in E.ON’s statement, is reported for every output.

Table A5: N-1 screening of decoded siting plans as a data-centre load is added at one bus: ten instances per row, every branch outaged in turn. “Secure” is the share of outages served with zero shedding; “EUE” the expected unserved energy over the outage set. Daggered rows: the intact power flow can no longer serve the added load at all, so both arms are compared on identical stressed conditions.

Added load	Secure fraction		EUE (MW)		EUE cut
	none	plan	none	plan	
<i>IEEE-57, 80 outages per case</i>					
50 MW	94.3%	96.1%	0.38	0.09	74.0%
100 MW [†]	85.7%	96.6%	1.43	0.09	92.8%
200 MW [†]	36.4%	96.1%	27.2	0.38	93.5%
350 MW [†]	0.0%	62.2%	171.2	8.5	95.4%
<i>IEEE-118, 186 outages per case</i>					
100 MW	93.6%	94.1%	1.95	1.62	17.0%
200 MW [†]	0.0%	93.6%	17.7	1.74	88.8%
350 MW [†]	0.0%	0.0%	160.5	29.3	81.9%

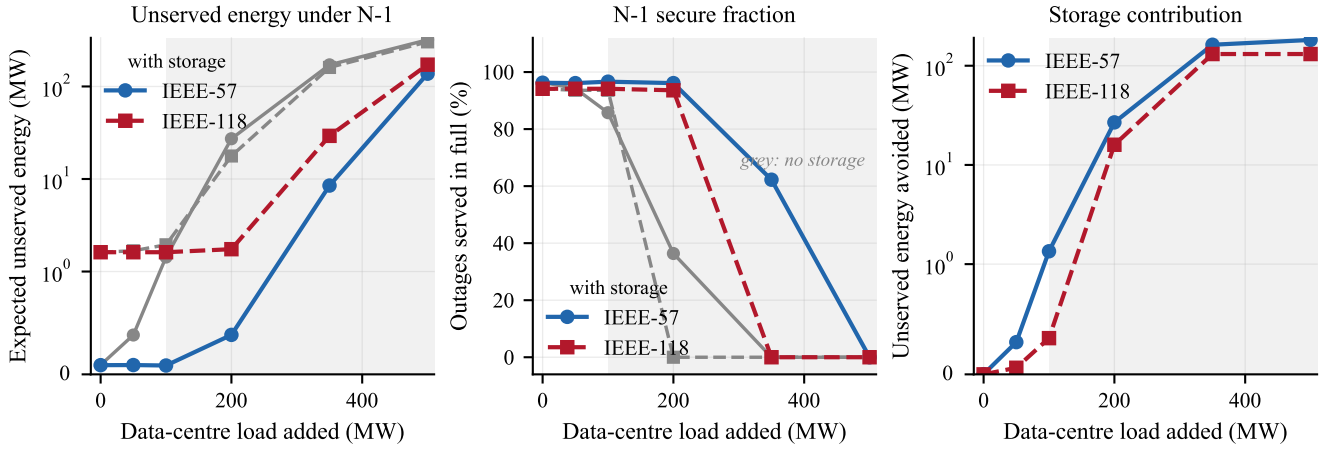


Figure A2: N-1 security of decoded plans as load is added at one bus. Coloured curves: network with the decoded plan; grey: same network and same outages without it. Shading marks the regime in which the intact power flow can no longer serve the added load. Axes are symmetric-log.

Extra decisions at zero qubit cost. Adding one islanding binary per candidate bus took the siting problem from $m = 40$ to $m = 44$, which the six-qubit register (capacity 45) absorbed without a new qubit; the solver reached the certified optimum on 60.9% of 64 extended instances, no worse than on the smaller problem. The same headroom argument is how conductor tiers and switch states will share a register with line additions in the PoC.

Appendix D. Repository map and reproduction

The public repository holds this proposal, its figures and their generating script, and the two code bases the PoC builds on, each with its own reproduction commands:

- `qgridx/`: the grid planning package (also on PyPI as `qgridx`), with `qgridx --list` printing one command per published table and figure, and two qBraid notebooks (`01_quickstart.ipynb`, `02_hardware.ipynb`). Result files: `results/doe_phase3/` (benchmark, scenarios, resources, N-1, microgrid), `results/hardware/` (Cepheus campaign).
- `genpce/`: the Qiskit-native PCE circuit-synthesis library. `experiments/e1_budget_curves.py` reproduces Table A2 and Figure 2b of the proposal; `e2_reward_robustness.py` reproduces Table A3; `tests/` holds the 26 unit tests that check the Walsh-Hadamard read-out against Qiskit’s Pauli expectation values.
- `proposal/`: this document and its sources.

Every experiment is seeded, resumable, and appends to CSV; the optimum of an instance is never used by any solver, only to report approximation ratios.